

Ahmad (Isa) Alsafwah

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EDUCATION

Boston University

Bachelor of Arts in Computer Science

Boston, MA

Dean's list, Sep. 2022 – May 2025

Rhodes College

Bachelor of Science in Computer Science

Memphis, TN

Aug. 2021 – May 2022 (Transferred)

EXPERIENCE

Full-Stack Developer Intern

May 2024-May 2025

Questrom School of Business, BU

- Conducted behavioral economics research focusing on designing market mechanisms to combat misinformation while preserving free speech. Contributed to presented findings at invited talks, including Google Monetization Policy (2025), Yale AI/ML Conf., WISE Conf., Columbia MarkTech, and MIT Decentralized AI Summit (2024).
- Designed and developed a human vs. human marketplace using React, Node.js, TailwindCSS, and MIT's Empirica library, incorporating multi-user games to simulate warranted markets for producers and consumers.
- Rearchitected backend logic to support multiple products per producer, replacing single-product assumptions; overhauled data models and implemented an interactive slider for dynamic product configuration.
- Translated collaboratively designed Figma designs into a responsive frontend using React and TailwindCSS, aligning UI behavior with user research and design goals.

PROJECTS

Multi-threaded Image Processing Server | C, POSIX Threads, Concurrency

Fall 2024

- Developed a multithreaded C server to process image transformations (rotate, blur, sharpen, edge detection) over TCP/IP in FIFO order.
- Implemented POSIX threads, semaphores, and configurable worker pools for concurrency control and robust queue management.
- Added queue-size limits and per-request timestamp logging to ensure scalability, safe resource sharing, and overload prevention.

MeowPlay | Javascript, Express, NodeJS, MongoDB

Full-stack Developer, Sep 2023

- Developed a full-stack e-commerce platform for cat products, implementing product listings, shopping cart, and secure checkout features.
- Integrated OAuth authentication, designed MongoDB schemas, and built backend logic with NodeJS/Express for reliable and secure performance.

Latent Semantic Analysis (LSA) Search Engine | Python, Flask, scikit-learn, NLP

Spring 2025

- Built a real-time semantic search engine over 18,000+ documents using TF-IDF vectorization and Latent Semantic Analysis.
- Reduced term-document matrix dimensionality by 90% using Truncated SVD, improving query processing speed by 40%.
- Implemented cosine similarity ranking to retrieve top 5 most semantically relevant results for each user query.
- Integrated Flask backend with HTML/JS frontend for responsive user interaction and real-time result delivery under 200ms latency.

Nutrisistant | Typescript/Javascript, Express, NodeJS, MongoDB

Backend Developer, Dec 2023

- Built a RESTful API using Express/NodeJS with MongoDB integration to manage user history and personalize content delivery.
- Integrated external APIs (CalorieNinjas, Spoonacular) for dynamic nutritional data retrieval and processing.

TECHNICAL SKILLS

Programming: Python, Java, C, Bash, JavaScript, Typescript, Assembly, HTML/CSS

Developer Tools: React, TailwindCSS, Next.js, ExpressJS, NodeJS, EJS, SQL, MongoDB, Shadcn, Bootstrap, Bulma, Railway, Git/Github, NumPy, Figma, Empirica

Relevant Coursework: Software Engineering, Full Stack Development, Algorithms, Databases, Design of Programming Languages, Embedded Systems, Information Security

Languages: English, Arabic, Spanish